Groups

$\mathbb{N}_0 = \{0, 1, 2, 3 \ldots\ldots\}$ (here $= \mathbb{N}$)

= natural numbers or non-negative integers

$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\ldots\}$

Integers

Made with Goodnotes

How to construct $\mathbb{Z}$ from $\mathbb{N}$?

We know that we can "add" natural numbers

$1+1=2$, $2+3=5$, ...

what does it mean to "add" two numbers?

Def (Binary operation)

Let $M \neq \emptyset$ be a set. We call a mapping

$$\circ : M \times M \longrightarrow M \qquad M \times M \ni (a,b) \longmapsto \underbrace{a \circ b}_{= \circ(a,b)} \in M,$$

a binary operation in M

$\circ =$ "$+$" $\quad \circ =$ "$\cdot$"

Made with Goodnotes

Remarks / examples

* "+" on $\mathbb{N}$ is a binary operation, as for all $a, b \in \mathbb{N}$ we have $+(a,b) = a+b \in \mathbb{N}$ e.g.
$+(2,3) = 2+3 = 5 = +(2,3)$

* Since the codomain is $\mathbb{N}$ again, we call this operation closed.

* $\mathbb{R}$ the real numbers, "+", "−", "·" are binary operations on $\mathbb{R}$. But $M = \{1,2,3\}$ "·" <u>not</u> a binary operation

Made with Goodnotes

* if $\circ(a,b) = \circ(b,a)$, we call $\circ$ commutative

$a + b = b + a$

in general $b - a \neq a - b$

Made with Goodnotes

The neutral or identity element

We consider $M = \mathbb{N}$, and $\circ = "+"$, Clearly we have

$$a = \circ(a, 0) = a + 0 = 0 + a = \circ(0, a) \quad \forall a \in \mathbb{N}$$

Similarly, for $\circ = "\cdot"$ and $M = \mathbb{R}$, we have

$$\circ(a, 1) = a \cdot 1 = 1 \cdot a = \circ(1, a) = a \quad \forall a \in \mathbb{R}$$

We say that 0 is the neutral element w.r.t. "+" and that 1 is the neutral element w.r.t. "$\cdot$"

Made with Goodnotes

We now consider the pairs $(\mathbb{N}, +)$, $(\mathbb{Z}, +)$

We know on $\mathbb{N}$ that $a + 0 = a \quad \forall a \in \mathbb{N}$

However, the problem (given $a \in \mathbb{N}$)

Find $b \in \mathbb{N}$ such that $a + b = 0$

We cannot solve in $\mathbb{N}$ (for $a \neq 0$).

We can solve in $\mathbb{Z}$, as we have the negative numbers

$$a + (-a) = 0 \quad \forall a \in \mathbb{Z}$$

GOAL: we would like to formalize this aspect

Made with Goodnotes

Group

Def: A group is a set $G \neq \emptyset$ equipped with a binary operation $\circ : G \times G \to G$ such that hold

(G1): $(a \circ b) \circ c = a \circ (b \circ c) \quad \forall a, b, c \in G$

associativity

(G2): there exists $e \in G$ such that

neutral element

$$a \circ e = e \circ a = a \quad \forall a \in G$$

(G3) For each $a \in G$ there exists $b \in G$ such that

inverse element

$$a \circ b = b \circ a = e$$

Made with Goodnotes

Remarks

* b in (G3) is called the inverse element of a with respect to "$\circ$". Usually we write $b = a^{-1}$ $(= -a)$

* if $a \circ b = b \circ a$ $\forall a, b \in G$ we call our binary operation commutative

* ex: show that a^{-1} is unique i.e.

$$\forall a \in G \quad \exists!\, a^{-1} \in G : a \circ a^{-1} = e = a^{-1} \circ a$$

Made with Goodnotes

Construction of $\mathbb{Z}$ using equivalence classes

$(\mathbb{N}, \text{"}+\text{"})$ has the property: $a, b \in \mathbb{N}$, $a+b \in \mathbb{N}$

$(*)$ $5 + a = 1$ obvious: $a = 1 - 5 = -4$

Question How to solve equation $(*)$?

How to define properly negative numbers?

Made with Goodnotes

$5 + a = 1 \qquad a = 1 - 5 = -4 \qquad (\overset{b}{1}, \overset{c}{5})$

$6 + a = 2 \qquad a = 2 - 6 = -4 \qquad (\overset{d}{2}, \overset{e}{6})$

$7 + a = 3 \qquad a = 3 - 7 = -4 \qquad (3, 7)$

$b + e = d + c \iff b - c = d - e$

R on $\mathbb{N} \times \mathbb{N}$: $(b, c) \sim (d, e) \iff b + e = d + c$

Made with Goodnotes

Show that R as defined above is an equivalence relation on $\mathbb{N} \times \mathbb{N}$

1) reflexive $(b, c) \sim (b, c)$ $b + c = b + c$

$\forall b, c \in \mathbb{N}$

2) symmetric $(b, c) \sim (d, e) \Leftrightarrow b + e = d + c$

$(d, e) \sim (b, c) \Leftrightarrow d + c = b + e$

3) Transitive $(b, c) \sim (d, e)$ and $(d, e) \sim (f, g)$

$b + e = d + c$ and $d + g = f + e$

$\rightarrow$ GOAL: $(b, c) \sim (f, g) \Leftrightarrow b + g = f + c$

Made with Goodnotes

TS Show $(b,c) \sim (f,g) \Leftrightarrow b+g = f+c$

Hy (1) $b+e = d+c$

(2) $d+g = f+e$

$(1) + f + g$

$b+e+f+g = d+c+f+g$

$(b+g) + \underline{\underline{\cancel{(e+f)}}} = (f+c) + \cancel{(d+g)}$

$\cancel{(2) = d+g}$

$b+g = f+c \qquad \square$

* We have used the <u>cancellation law</u> on $\mathbb{N}$

For any $x, y, z \in \mathbb{N}$, if $x+z = y+z$, then $x = y$

Made with Goodnotes

Consequence: R as defined above <u>IS</u> an equivalence relation on $\mathbb{N} \times \mathbb{N}$

* We will associate with the equivalence classe "our" integers: $\mathbb{Z} := \mathbb{N}/R$

* How to define "+" on equivalence classes?

Made with Goodnotes

Addition on $\mathbb{Z}$

Assume $[(a,b)] \in \mathbb{Z}$ and $[(c,d)] \in \mathbb{Z}$

We would like to define addition on $\mathbb{Z}$

$$[(a,b)] +_R [(c,d)] = [(a+c,\ b+d)]$$

Made with Goodnotes